Weathering, Erosion, Deposition, and Landscapes

Weathering – the breakdown of rocks into smaller pieces, called sediments.

Erosion – the process where the sediments are transported by wind, gravity, glaciers, man, and running water.

Deposition – the process whereby these sediments are released by their transporting agents (dropped).

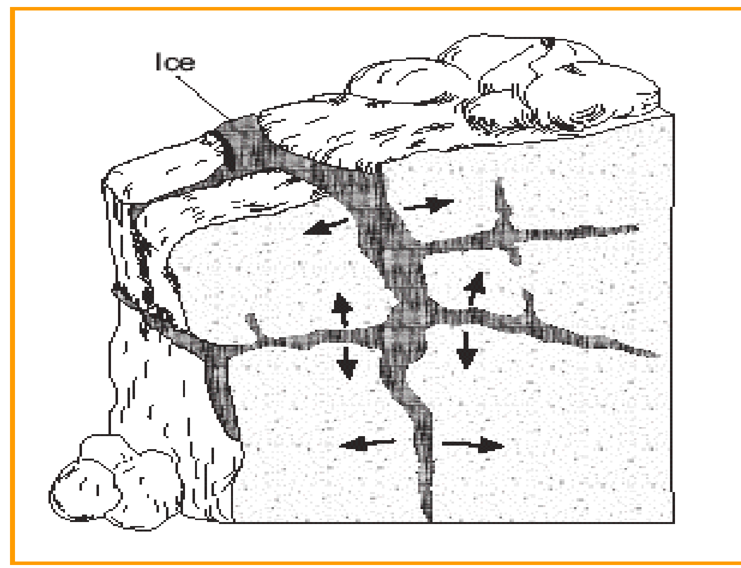
Weathering breaks down the rocks, erosion moves the particles, and deposition drops the sediments in another location.

There are two primary types of weathering:
Chemical and Physical

1. Physical Weathering – the breakdown of rocks and minerals into smaller pieces without a change in chemical composition.

Root/Plant Wedging/Action

Ice/Frost Wedging/Action



Exfoliation and ***Abrasion*** are also types of physical weathering.

2. Chemical Weathering – the breakdown of rocks and minerals into smaller pieces by chemical action. The rocks breaks down at the same time as it changes chemical composition. The end result is different from the original rock. **There are 3 types of chemical weathering:**

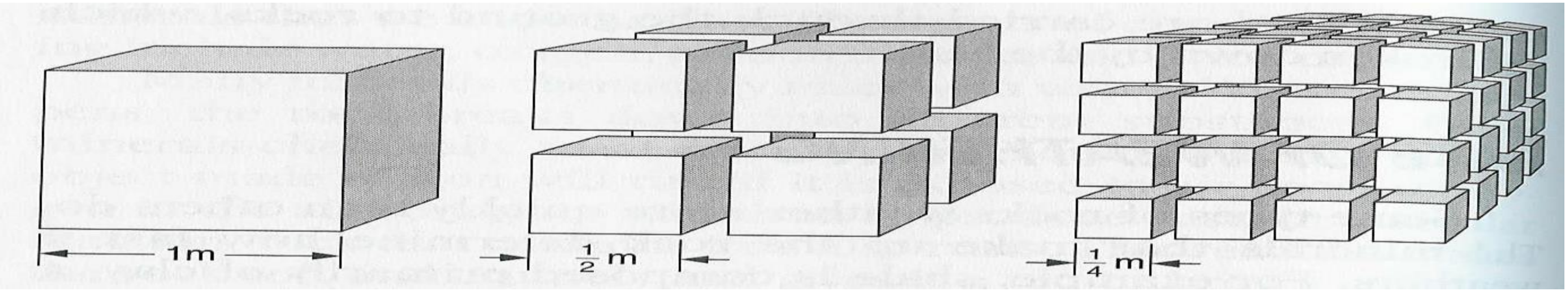
1. Oxidation – oxygen combines with the elements in the rock and it reacts. This the scientific name for rust. →

2. Hydration – water can dissolve away many earth materials, including certain rocks.

3. Carbonation – carbon dioxide dissolves in water to form carbonic acid. This makes acid rain which chemically weathers (dissolves) rocks. Other acids also combine with water to make acid rain.

There are 4 factors that effect the rate of weathering:

1. ***Surface Area*** (exposure) - Exposing more surface area will increase the rate of weathering.



2. ***Particle Size*** – Larger particles weather slower and smaller particles weather at a faster rate.

3. ***Chemical Composition*** (what a rock is made of) – Certain rocks and minerals are naturally weaker than others, while others are more resistant (stronger).

4. ***Climate*** – Warmer, moister climates have the most weathering. Heat & Water speed up all chemical reactions. This is the most important factor in weathering.

Soil forms from the weathering of the rock below it. The solid rock below is called ***Bedrock***. The rock is exposed to wind, rain etc… The rock breaks down over time to form soil. Soil has different layers called ***Soil Horizons.***

O- Horizon = the very thin surface covering (not really a layer)
A – Horizon (TOPSOIL) = dark surface soil that contains a lot of living material and dead plant/animal remains (humus). This is the layer with all of the nutrients needed to grow plants.
B-Horizon (SUBSOIL) = lighter colored soil with less nutrients and more clay
C-Horizon (REGOLITH) = larger rock fragments that sit on top of the unweathered bedrock

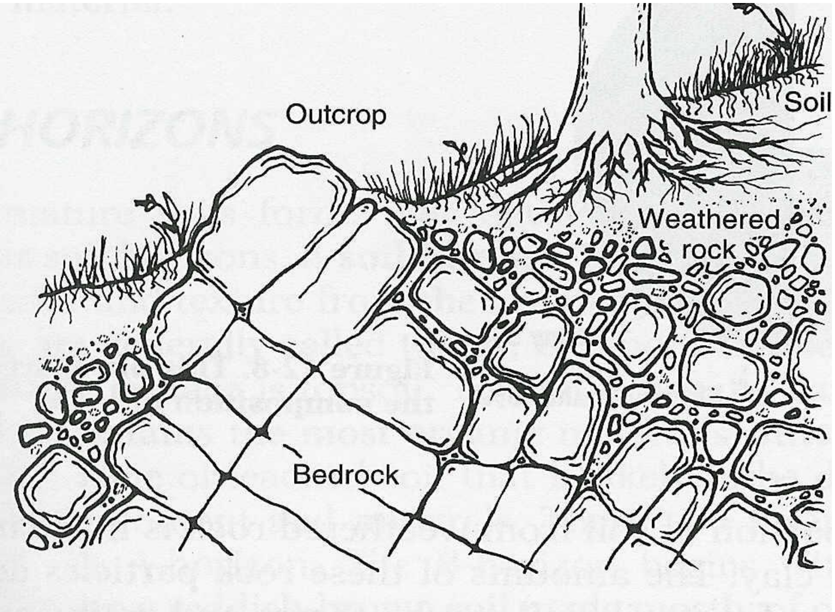


Figure 12-6. Rocks gradually weather over time to form soil.

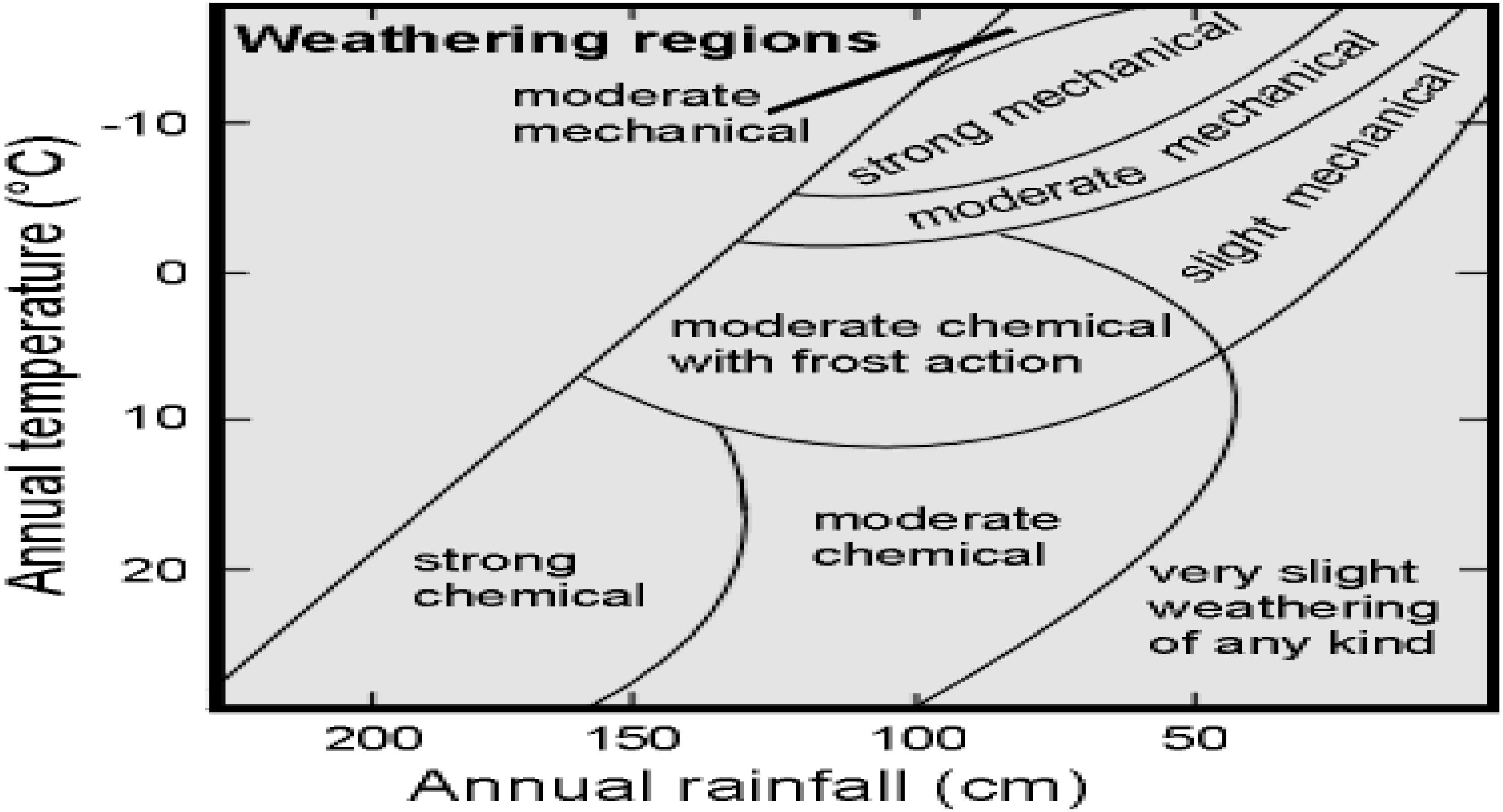
There are 2 types of soil:

1. Transported
2. Residual

Transported soils are the most common throughout New York State!!!!!!!!

Transported Soils – soils that formed in one place and were transported to their present location by glaciers. You can tell when the soil does not chemically match the bedrock below it.

Residual Soils – soils that are located above the rocks that they formed from. In other words, the soil chemically matches the bedrock below it, because it is a product of that rock's weathering over time.



What type of weathering occurs in an area that has an average of 165 cm of rain each year and an average temperature of 18°C? ___________________________

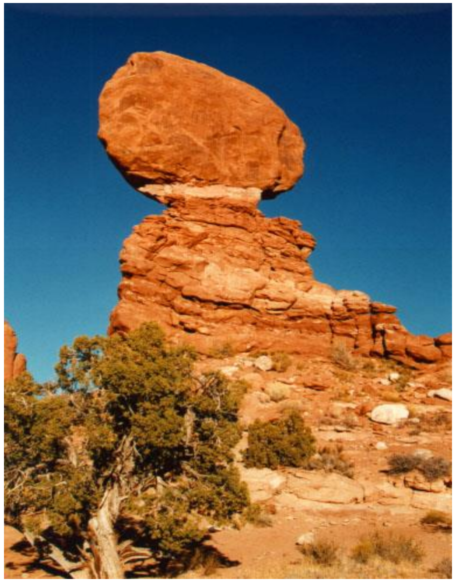
What type of weathering occurs in an area that has about 140 cm of precipitation each year and an average temperature of 5°C? ____________________________

Erosion (transport)

There are 5 main agents of erosion:

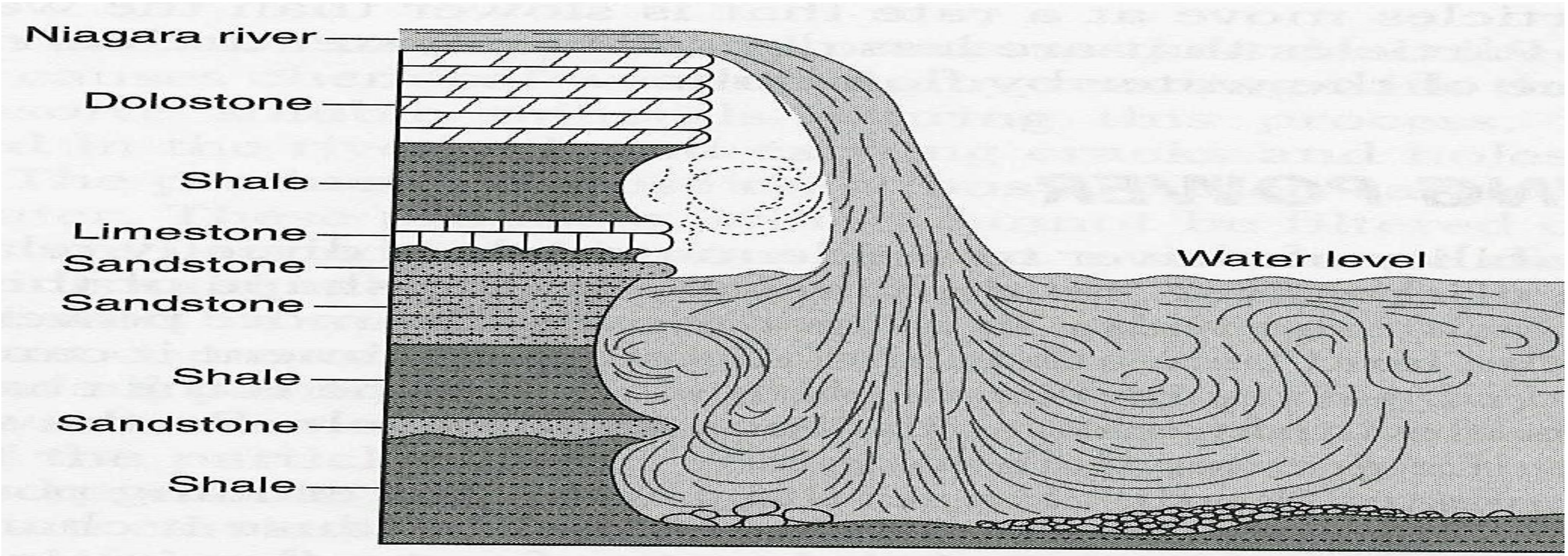
1. *Running Water*
2. Glaciers
3. Wind
4. Gravity
5. Man

Weathering has to happen before erosion. The rocks have to be broken into smaller sediments before they can be eroded away.



← Wind Erosion

Glacier →



Stream *erosion* is the greatest at waterfalls.
Erosion at waterfalls is called ***undermining***.

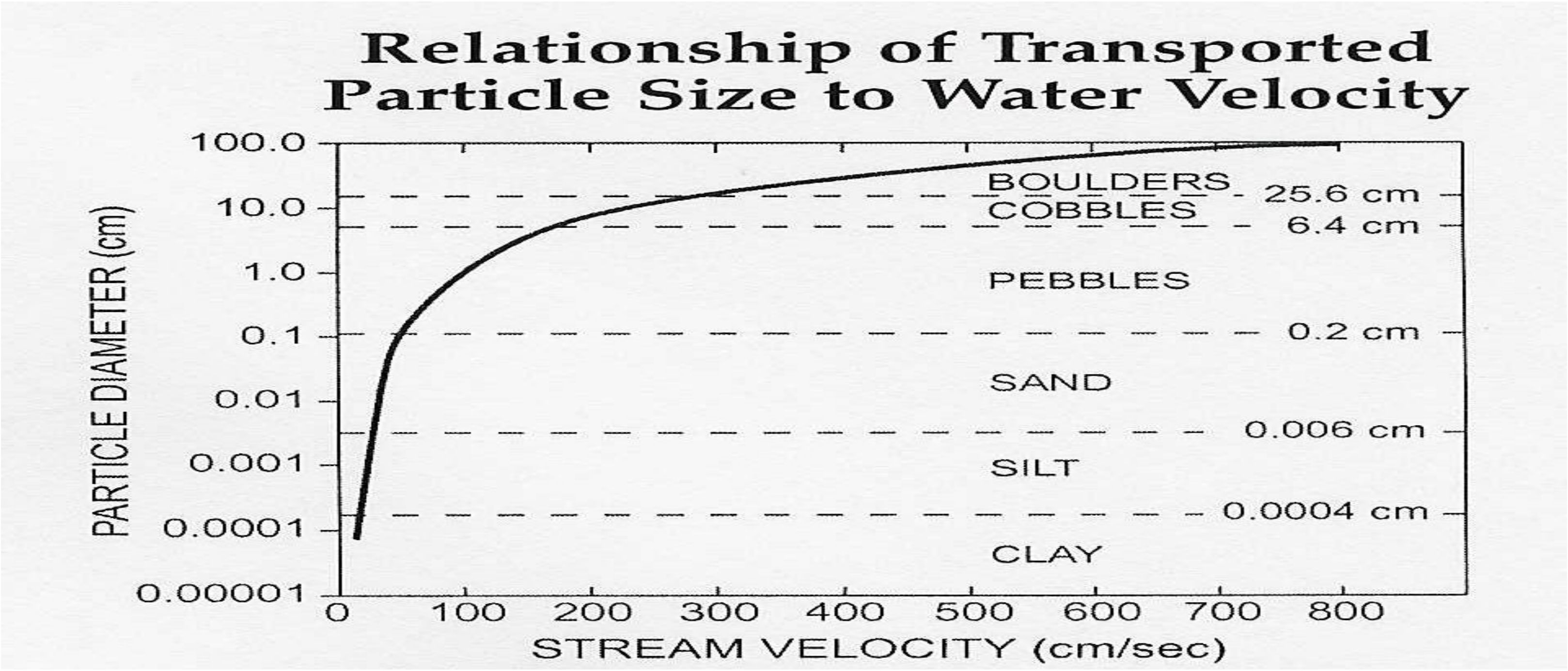
Which rock layer appears to be the least resistant (weakest)? ________

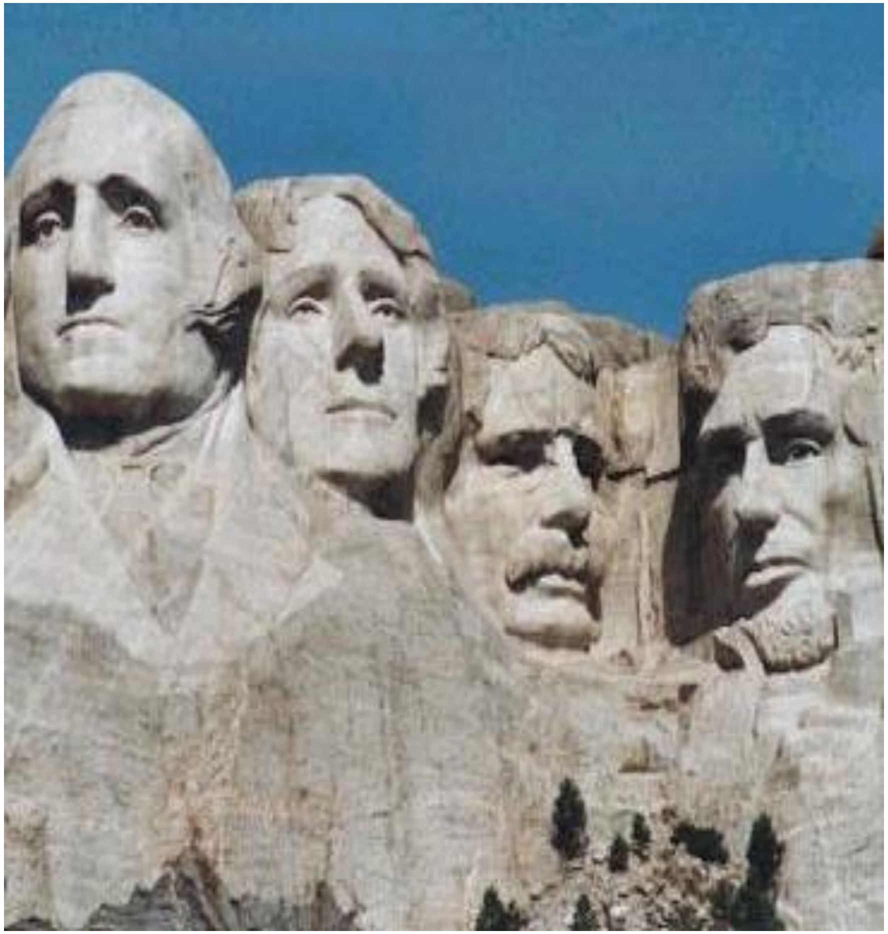
Which rock layer appears to be the most resistant (strongest)? ________

Resistant rocks usually form steep cliffs and waterfalls, by sticking out further than the lower layers.

There are 4 basic products of weathering, that can be eroded:

1. Soils
2. Solid Sediments (boulders, cobbles, pebbles, sand, silt)
3. Colloids/Clay Particles (not visible to your eye)
4. Ions (very small electrically charged particles)





Mount Rushmore

It will not be there forever!!

There is a pile of weathered material at the bottom. It is slowly being eroded down hill by gravity.

You can identify which agent of erosion transported each sediment by looking at a few characteristics:

Running Water – sediments that have been transported through running water appear **rounded** and **smooth** and are deposited in **sorted** piles.

Glaciers – sediments that have been transported by glaciers appear **scratched, grooved**, and are deposited in **completely unsorted** piles, because they were dropped during melting. Also, **boulders** can only be transported by glaciers.

Wind - sediments that have been transported by wind are appear **pitted** (random holes) and **frosted** (glazed look) and are deposited in **sorted** piles. Only very **small** particles can be transported by wind.

Gravity – sediments that are transported by gravity are found in piles at the **bottom of cliffs** or steep slopes. They appear **angular** and **unsorted.**

Extreme Wind Erosion

Melbourne dust storm, 1983

The rocks to the right were transported by **running water**. How can you tell? ________________

Glacial Striations

← (scratches)

Grain Size
Udden-Wentworth Scale

Millimeters (mm)		Micrometers (µm)	Phi (ϕ)	Wentworth size class		Rock type
	4096		-12.0	Boulder	Gravel	Conglomerate/ Breccia
	256		-8.0	Cobble		
	64		-6.0	Pebble		
	4		-2.0	Granule		
	2.00		-1.0	Very coarse sand	Sand	Sandstone
	1.00		0.0	Coarse sand		
1/2	0.50	500	1.0	Medium sand		
1/4	0.25	250	2.0	Fine sand		
1/8	0.125	125	3.0	Very fine sand		
1/16	0.0625	63	4.0	Coarse silt	Silt	Siltstone
1/32	0.031	31	5.0	Medium silt		
1/64	0.0156	15.6	6.0	Fine silt		
1/128	0.0078	7.8	7.0	Very fine silt		
1/256	0.0039	3.9	8.0	Clay	Mud	Claystone
	0.00006	0.06	14.0			

Boulder > 256 mm (-8 to -12 ϕ)

Wentworth Size Class	mm scale		phi scale
Pebbles	256 to 4		-8 to -2
Gravel	4 to 2		-2 to -1
Very coarse and Coarse sand	2 to 0.5		-1 to 1
Medium sand	0.5 to 0.25		1 to 2
Fine and Very fine sand	0.25 to 0.06		2 to 4
Silt	0.06 to 0.004		4 to 8
Clay	< 0.004		> 8.00

Running water can transport sediment in three ways:

1. *Solution* – the smallest particles of weathering are dissolved in the water and they are transported in a solution.
2. *Suspension* – clay sized/colloids are carried along with the water molecules during erosion. They are neither at the bottom or on the top. They are suspended in the middle of the running water.
3. *Saltation* – solid sediments bounce along the bottom of a river stream because they are more dense.
4. Traction -Traction is the geologic process whereby a current transports larger, heavier rocks by rolling or sliding them along the bottom.

Erosion vocabulary and facts:

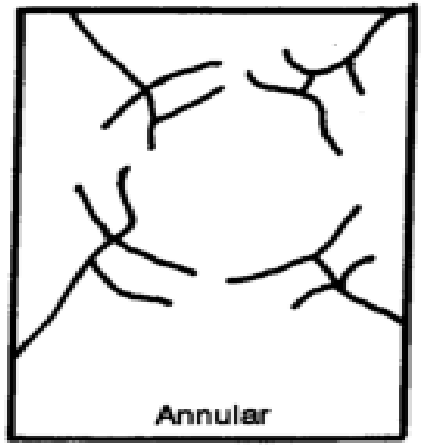
The sediments that are being transported by the river/stream are traveling a little bit slower than the water. This is because of friction.

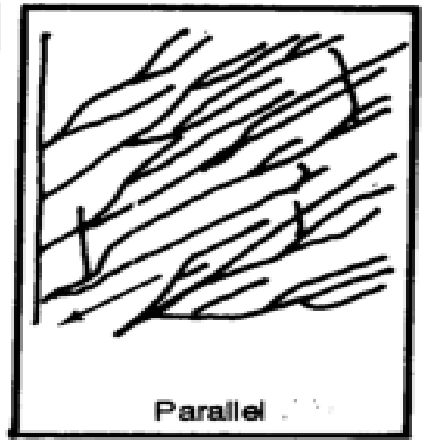
Stream/River Bed – the bottom of a stream or river.

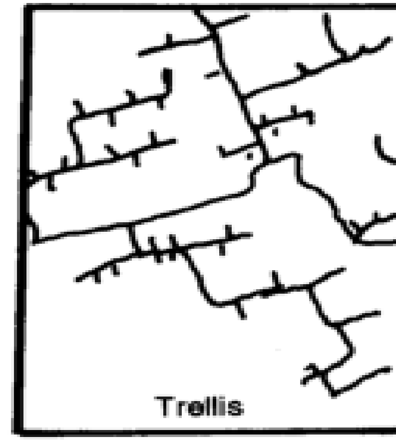
Bed Load – the material being transported along the bottom of a river/stream (rocks and pebbles).

Downcutting – when weathering and erosion, along with the running water, cause the stream/river to become wider and deeper over time. Younger streams/rivers are more shallow and narrow. Older rivers/streams are wider and much deeper.

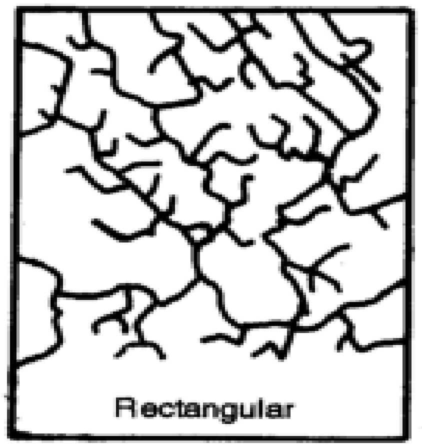
This is how water drains off of mountains that are made of many different rocks.

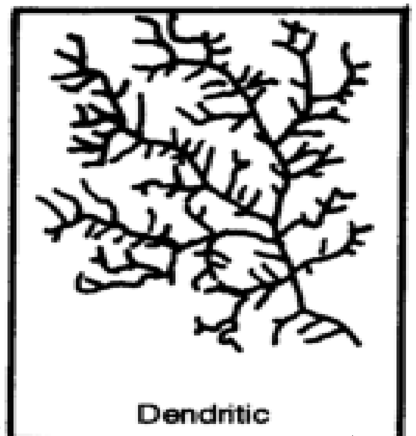


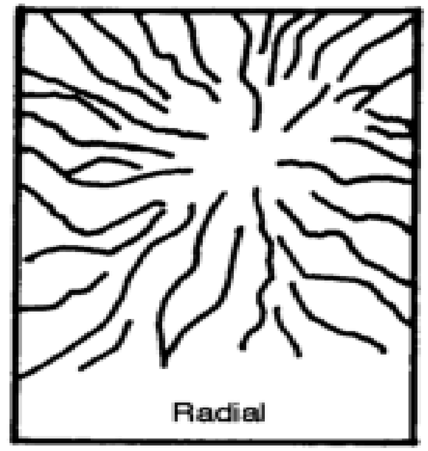


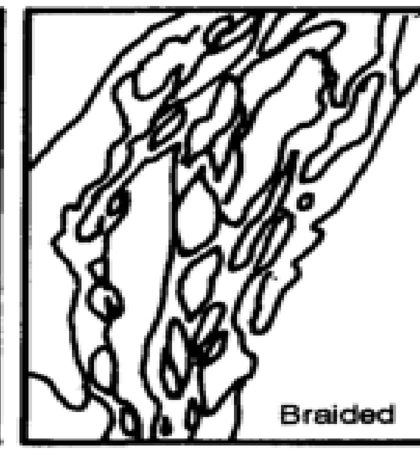


Stream Drainage Patterns









This is how water drains in faulted areas or areas that are made of many different rocks.

This is how water drains in flat areas & areas that are made of the same rock types.

This is how water drains from mountains and volcanoes that are made of the same rocks.

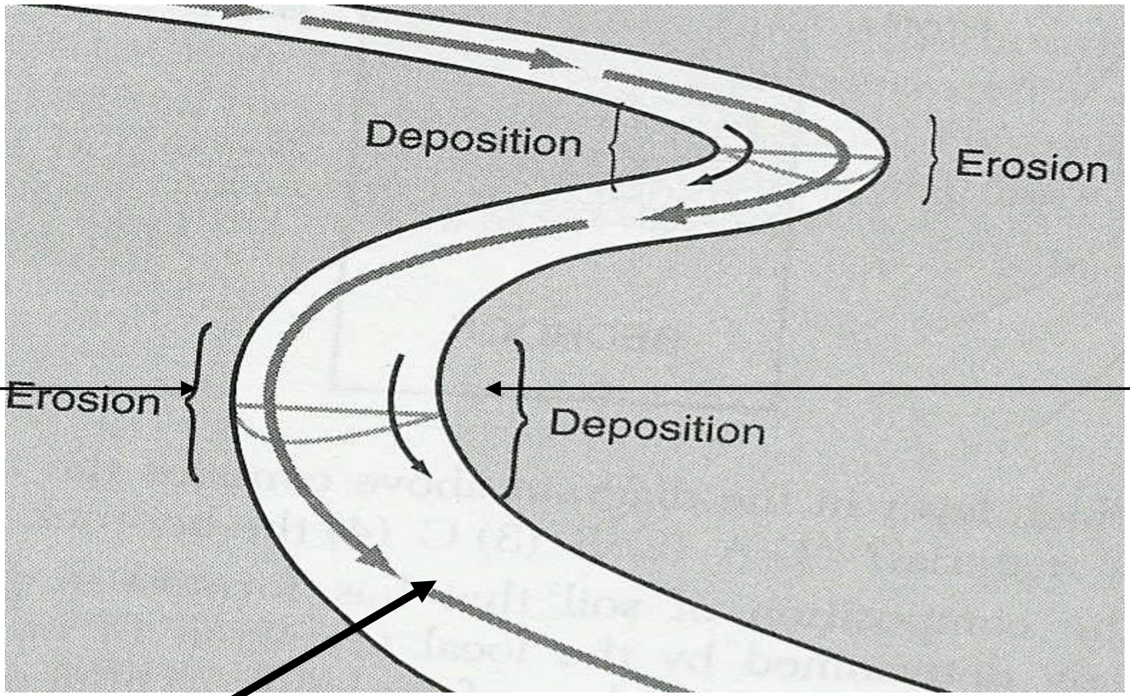
Factors that influence erosional rates (speed) in running water and glacial ice:

1. **Slope (gradient) of the land** – as slope increases, the water velocity increases, the particle size that the water can carry also increases, therefore the amount of erosion increases.
2. **Volume (size of the water or glacier) –** as the volume of the water or glacier increases, their velocities increase, the particle size that they can carry also increases, therefore the amount of erosion increases.
3. **Position within the running water** – Water is traveling faster around the outside of turns, therefore that is where more erosion occurs. Water is traveling slower on the inside of turns, therefore deposition occurs on the inside.

(see diagram on next slide)

Meandering (Curving) River/Stream

Erosion happens on the outside of turns.



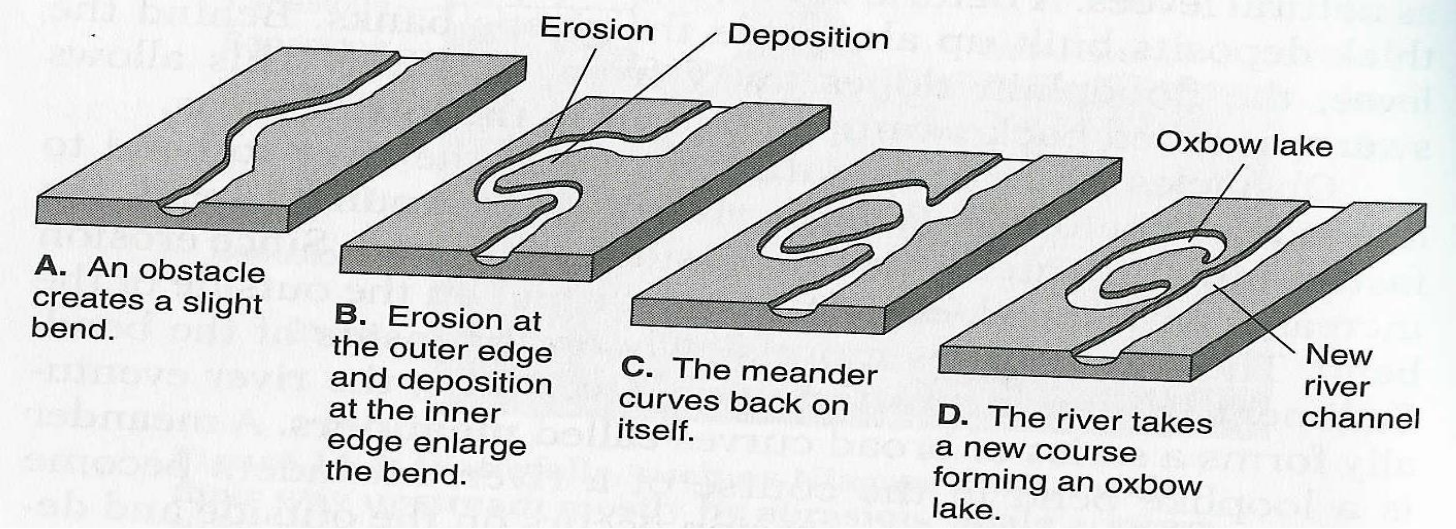
Deposition happens on the inside of turns.

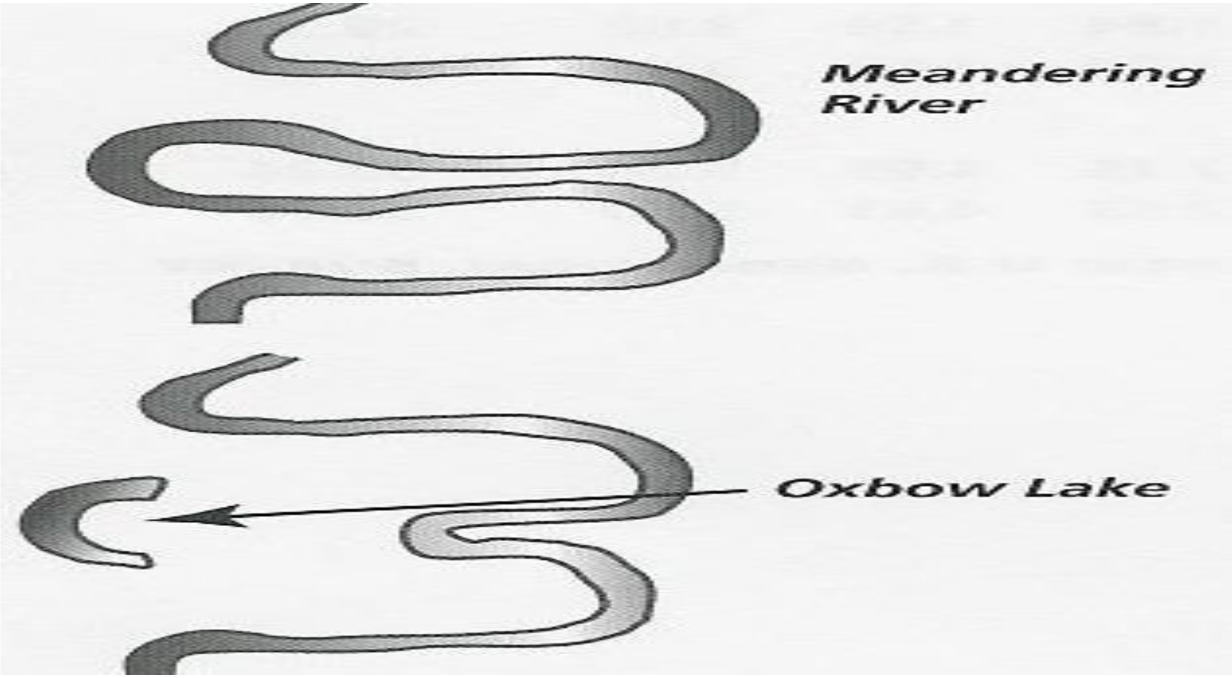
Sediments are traveling the fastest in the center directly below the surface.

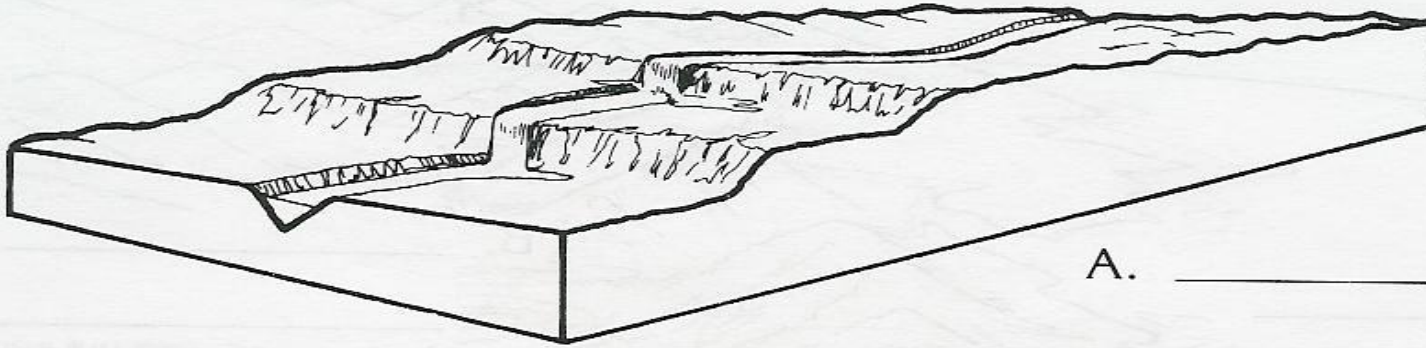
Straight Flowing River/Stream

Running Water

O x b o w L a k e

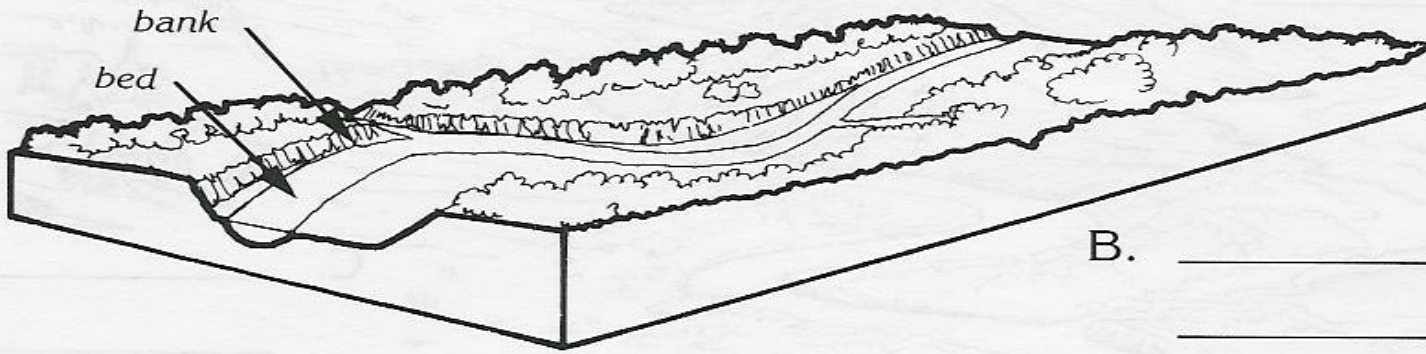

Erosion
Deposition
Oxbow lake
A. An obstacle creates a slight bend.
B. Erosion at the outer edge and deposition at the inner edge enlarge the bend.
C. The meander curves back on itself.
D. The river takes a new course forming an oxbow lake.
New river channel


Meandering River
Oxbow Lake



A. ______________________

____



B. ______________________

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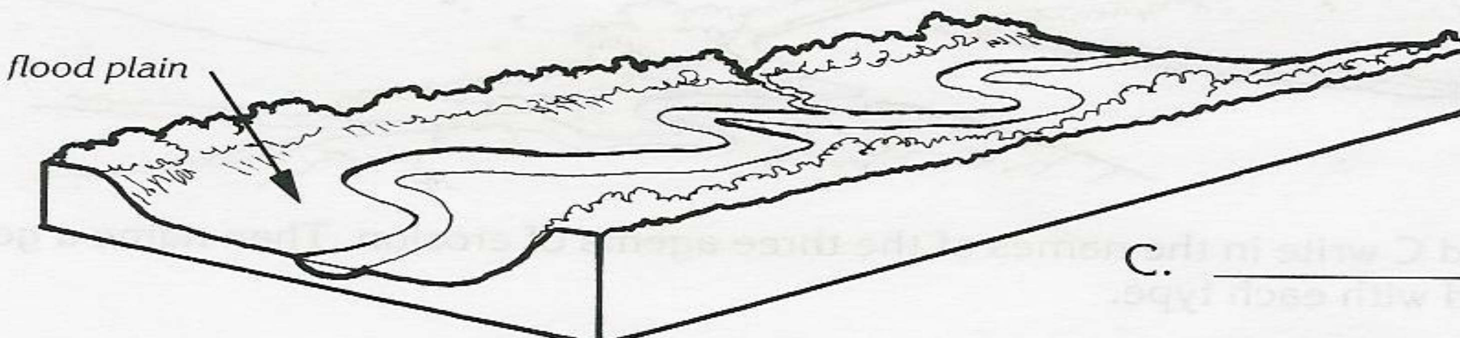


C. ______________________

____

There are 5 ways that man can cause erosion:

1. **Forestry** – all vegetation of removed, and without roots, the soil will erode away.
2. **Strip Mining –** removing rock cover to get to the resources below, which causes the loose sediments to erode away.
3. **Construction** – the clearing of land to build buildings/houses also causes all loose soil to erode away.
4. **Improper Farming** – not plowing the land at right angles to slopes causes soil to erode away.
5. **Salting Highways** – the salt is washed off the road to the sides, where it prevents vegetative growth along the sides.

Deposition – the process where sediments are released/dropped by their agent of erosion.

Most deposition happens in standing/still bodies of water (oceans/lakes).

Deposition is caused by the slowing down (loss of kinetic energy) of the agent of erosion.

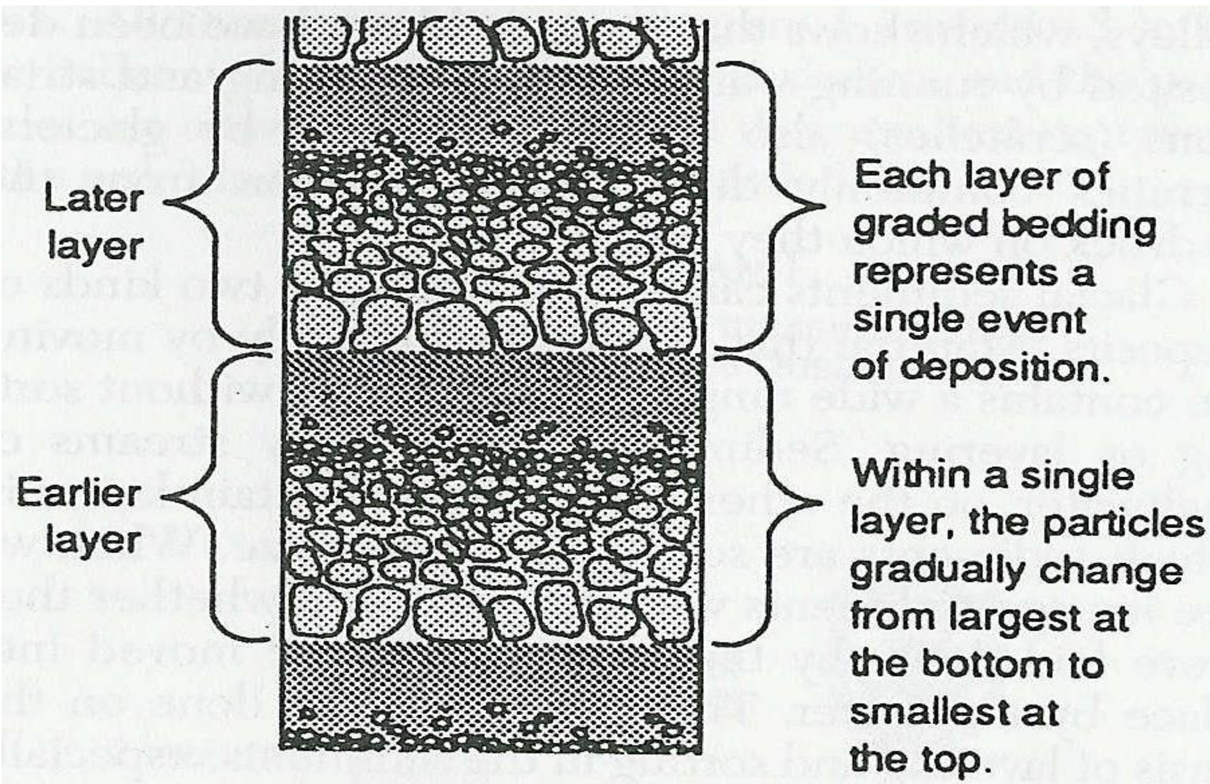
There are 3 factors that influence the rate of sediment deposition:

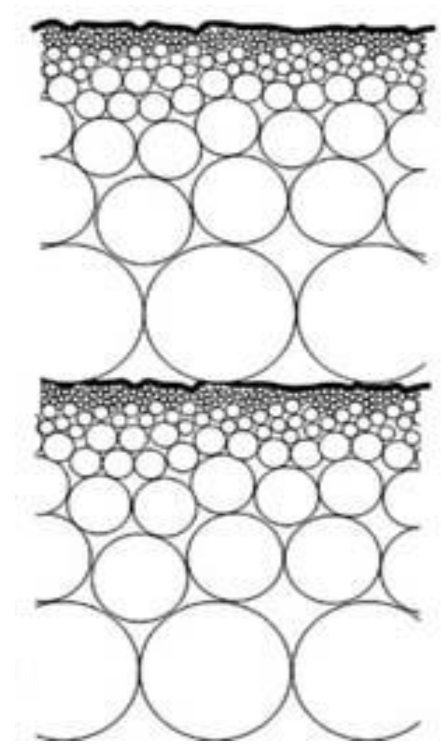
1. **Sediment size** –

2. **Sediment shape** –

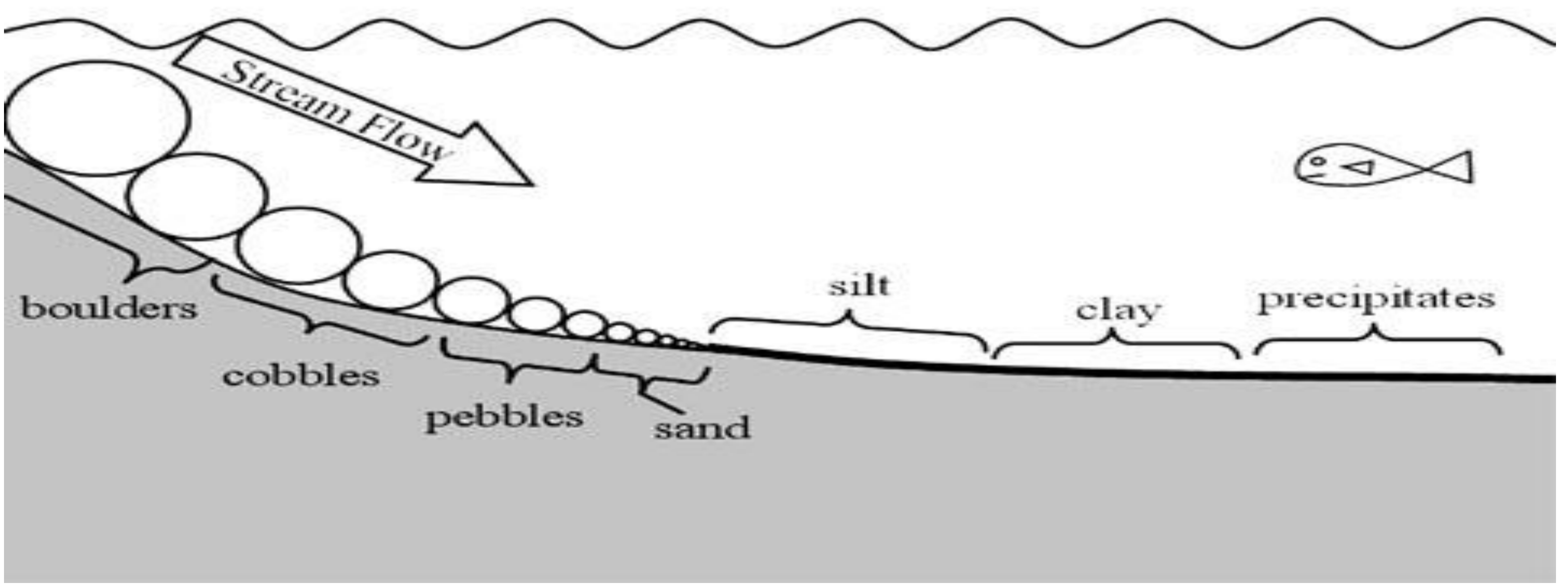
3. **Sediment density -**

Graded Bedding/Vertical Sorting – a situation where larger particles settle on the bottom and smaller particles settle towards the top. This happens naturally when a fast moving river/stream meets a large standing body of water. This happens because the velocity of the water decreases very quickly. (A waterfall emptying into a lake)





Horizontal Sorting – a situation where moving water enters a larger, still body of water slowly, and causes the larger particles to be deposited closer to the shoreline. Particle size decreases as you move away from the shore.



FACTORS THAT AFFECT POROSITY

PRIMARY

- Particle sphericity and angularity
- Packing
- Sorting (variable grain sizes)

Porosity Calculations - Uniform Spheres

- Bulk volume = $(2r)^3 = 8r^3$
- Matrix volume = $$\frac{4 \pi r^3}{3}$$
- Pore volume = bulk volume - matrix volume

$$\text{Porosity} = \frac{\text{Pore Volume}}{\text{Bulk Volume}}$$

$$= \frac{\text{Bulk Volume} - \text{Matrix Volume}}{\text{Bulk Volume}}$$

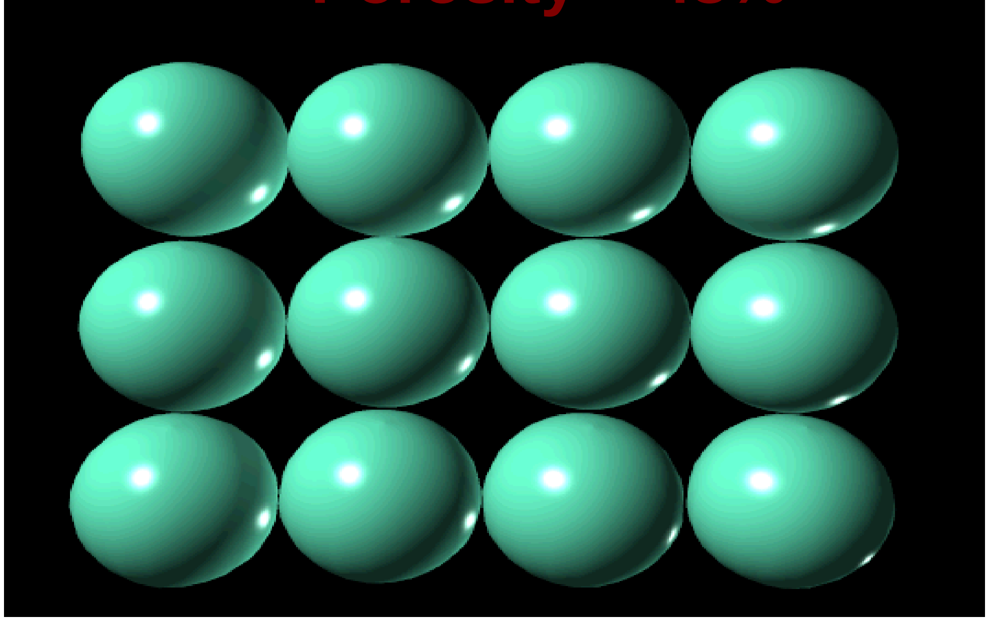
$$= \frac{8\,r^3 - 4/3\,\pi\,r^3}{8\,r^3} = 1 - \frac{\pi}{2\,(3)} = 47.6\%$$

Porosity Calculations - Uniform Spheres

- Bulk volume = $(2r)^3 = 8r^3$
- Matrix volume = $\dfrac{4\pi r^3}{3}$
- Pore volume = bulk volume - matrix volume

Porosity = 48%

CUBIC PACKING OF SPHERES



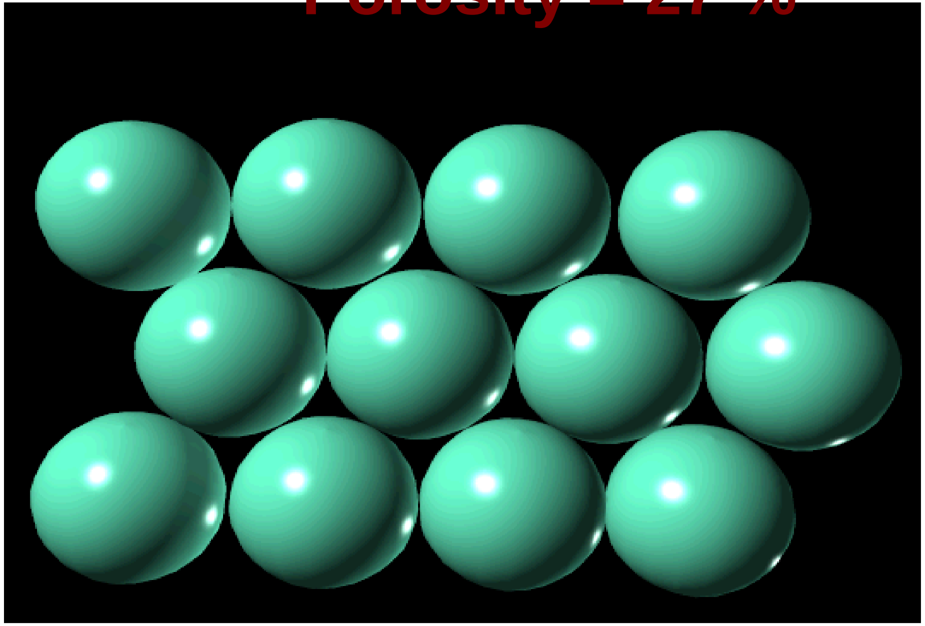
$$\text{Porosity} = \frac{\text{Pore Volume}}{\text{Bulk Volume}}$$

$$= \frac{\text{Bulk Volume} - \text{Matrix Volume}}{\text{Bulk Volume}}$$

$$= \frac{8\,r^3 - 4/3\,\pi\,r^3}{8\,r^3} = 1 - \frac{\pi}{2\left(3\right)} = 47.6\%$$

Porosity = 27 %

RHOMBIC PACKING OF SPHERES



FACTORS THAT AFFECT POROSITY

PRIMARY

- Particle sphericity and angularity
- Packing
- Sorting (variable grain sizes)

SECONDARY (DIAGENETIC)

- Cementing materials
- Overburden stress (compaction)
- Vugs, dissolution, and fractures

Packing of Two Sizes of Spheres

Porosity = 14%